

Dao Xuan Tan

Backend Engineer (Python)

📍 Ho Chi Minh, Vietnam

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🌐 Đào Xuân Tân

Summary

Backend engineer focused on fast, reliable Python services and tooling, FastAPI, PostgreSQL, Kafka, and Redis on Docker and Kubernetes, with the depth to chase problems down to memory and concurrency at the OS level.

Experience

Pi Associates, Backend Engineer

Ho Chi Minh

Python, PostgreSQL, Kafka, Redis, Docker, Kubernetes, GCP, GitHub Actions

Jan 2026 – present

6 months

- Main developer responsible for an internal **Python** backtesting package, used by about 20 researchers and engineers running **tens of thousands** of backtests a day
- Diagnosed and fixed a critical **memory leak**, cutting peak memory from **200+ GB to under 20 GB**
- Cut backtest runtime from **minutes to under 10 seconds** on **1-5 GB** of data per run through **parallelism**, vectorization, and algorithmic improvements
- Designed the **PostgreSQL** schema, **Redis** caching, and a **Kafka** producer feeding signals to downstream consumers
- Ran backtests as Dockerized **Kubernetes** jobs across **GCP** and on-prem, with **GitHub Actions** CI/CD and Grafana monitoring
- Improved reliability with better error handling and input validation, cutting failed runs and manual debugging

AcamarAI, Backend Engineer

Ho Chi Minh

Python, FastAPI, asyncio, PostgreSQL, Docker

June 2025 – Jan 2026

8 months

- Built **async FastAPI** services with full type checking and **Pydantic** validation, serving internal **REST APIs** through an OpenAI-compatible interface to Fortune 500 beta users
- Wrote heavily **asynchronous** backends, non-blocking endpoints, concurrent I/O, task orchestration, and streamed responses
- Owned **3** production features from requirements through rollout, out of **14** core features across the product
- Deployed self-hosted inference (vLLM) behind **FastAPI** and built evaluation and QA harnesses

Skills

Python: asyncio, multiprocessing and GIL-aware concurrency, memory profiling and leak debugging, typing (mypy, pyright, Pydantic), packaging (uv)

Backend & Data: FastAPI, REST and async APIs, PostgreSQL (SQLAlchemy and raw SQL), Redis, Kafka, data pipelines

Infra & DevOps: Docker, Kubernetes, GCP, on-premise, GitHub Actions CI/CD, Grafana

Testing: pytest (unit, integration, fixtures, mocking)

Other: LLM serving (vLLM), RAG and prompt engineering, PyTorch

Languages: Vietnamese (Native), English (IELTS 6.5)

Education

Ho Chi Minh University of Science, Mathematics and Computer Science, Data Science track

Ho Chi Minh City

Oct 2022 – Sept 2026

- GPA: 8.74/10.0 (3.74/4.0)